

Foundation (Jalapeno)

Q1. If $f(x) = x + 2$ and $g(x) = x - 1$, what is $(f + g)(x)$?

- (A) $2x + 1$
- (B) $2x - 1$
- (C) $x + 1$
- (D) $x - 1$
- (E) x

Q2. If $f(x) = x^2$ and $g(x) = x - 3$, what is $(f - g)(x)$?

- (A) $x^2 - x + 3$
- (B) $x^2 - x - 3$
- (C) $x^2 + x - 3$
- (D) $x^2 + x + 3$
- (E) $x^2 - 3$

Q3. If $f(x) = x^2$ and $g(x) = x + 2$, what is $(f \cdot g)(x)$?

- (A) $x^3 + 2x^2$
- (B) $x^3 - 2x^2$
- (C) $x^2 + 2x$
- (D) $x^3 + 2x$
- (E) $x^2 + 2x^2$

Q4. If $f(x) = x + 1$ and $g(x) = x - 2$, what is $(f \div g)(x)$?

- (A) $\frac{x+1}{x-2}$
- (B) $\frac{x+2}{x-1}$
- (C) $\frac{x-1}{x+2}$
- (D) $\frac{x-2}{x+1}$
- (E) $\frac{x+2}{x+1}$

Q5. What is the domain of $(f + g)(x)$ if $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-1}$?

- (A) $(-\infty, -1) \cup (-1, \infty)$
- (B) $(-\infty, 0) \cup (0, 1) \cup (1, \infty)$
- (C) $(-\infty, 0) \cup (0, \infty)$
- (D) $(-\infty, 1) \cup (1, \infty)$
- (E) $(-\infty, 0) \cup (0, 1)$

Q6. If $f(x) = x^2 - 4$ and $g(x) = x + 2$, what is $(f \circ g)(x)$?

- (A) $(x + 2)^2 - 4$
- (B) $x^2 + 4x + 4$
- (C) $x^2 - 4x - 4$
- (D) $x^2 + 2x - 4$
- (E) $x^2 - 2x - 4$

Q7. If $f(x) = x - 3$ and $g(x) = x^2 + 1$, what is $(f \cdot g)(x)$?

- (A) $x^3 - 3x^2 + x - 3$
- (B) $x^3 + 3x^2 + x + 3$
- (C) $x^2 - 3x - 3$
- (D) $x^2 + 3x - 3$
- (E) $x^3 - 3x^2 + x + 3$

Q8. What is the domain of $(f \div g)(x)$ if $f(x) = x^2 - 4$ and $g(x) = x + 2$?

- (A) $(-\infty, -2) \cup (-2, \infty)$
- (B) $(-\infty, 2) \cup (2, \infty)$
- (C) $(-\infty, 0) \cup (0, \infty)$
- (D) $(-\infty, -2)$
- (E) $(-2, \infty)$

Open Ended Questions (FRQ)

Q9. If $f(x) = x^2 - 2x$ and $g(x) = x + 1$, find $(f + g)(x)$ and $(f \cdot g)(x)$. Simplify your answers.

Q10. If $f(x) = \frac{1}{x-1}$ and $g(x) = x^2 + 1$, find $(f \div g)(x)$. Determine the domain of $(f \div g)(x)$.

Chef's Tips

- Tip 1: Use the given formulas to simplify the expressions
- Tip 2: Be sure to determine the domains of the functions
- Tip 3: Simplify all resulting expressions